What is the Windows Subsystem for Linux?

05/21/2025

Windows Subsystem for Linux (WSL) is a feature of Windows that allows you to run a Linux environment on your Windows machine, without the need for a separate virtual machine or dual booting. WSL is designed to provide a seamless and productive experience for developers who want to use both Windows and Linux at the same time.

- Use WSL to install and run various Linux distributions, such as Ubuntu, Debian, Kali, and more. Install Linux distributions and receive automatic updates from the Microsoft Store, import Linux distributions not available in the Microsoft Store, or build your own custom Linux distribution.
- Store files in an isolated Linux file system, specific to the installed distribution.
- Run command-line tools, such as BASH.
- Run common BASH command-line tools such as `grep`, `sed`, `awk`, or other ELF-64 binaries.
- Run Bash scripts and GNU/Linux command-line applications including:
  - Tools: vim, emacs, tmux
  - Languages: NodeJS, JavaScript, Python, Ruby, C/C++, C# & F#, Rust, Go, etc.
  - Services: SSHD, MySQL, Apache, lighttpd, MongoDB, PostgreSQL.
- Install additional software using your own GNU/Linux distribution package manager.
- Invoke Windows applications using a Unix-like command-line shell.
- Invoke GNU/Linux applications on Windows.
- Run GNU/Linux graphical applications integrated directly to your Windows desktop
- Use your device GPU to accelerate Machine Learning workloads running on Linux.

WSL is an Open Source tool with source code available for download and contributions:

- Learn more about WSL open source components
- WSL Open Source docs site: wsl.dev
- WSL repository on GitHub: github.com/Microsoft/wsl

Get started using WSL:

Install WSL

https://www.youtube-nocookie.com/embed/48k317kOxqg

What is WSL 2?

WSL 2 is the default distro type when installing a Linux distribution. WSL 2 uses virtualization technology to run a Linux kernel inside of a lightweight utility virtual machine (VM). Linux distributions run as isolated containers inside of the WSL 2 managed VM. Linux distributions running via WSL 2 will share the same network namespace, device tree (other than `/dev/pts`), CPU/Kernel/Memory/Swap, `/init` binary, but have their own PID namespace, Mount namespace, User namespace, Cgroup namespace, and `init` process.

WSL 2 **increases file system performance** and adds **full system call compatibility** in comparison to the WSL 1 architecture. Learn more about how WSL 1 and WSL 2 compare.

Individual Linux distributions can be run with either the WSL 1 or WSL 2 architecture. Each distribution can be upgraded or downgraded at any time and you can run WSL 1 and WSL 2 distributions side by side. See the Set WSL version command.
https://www.youtube-nocookie.com/embed/MrZolfGm8Zk

Microsoft Loves Linux

Learn more about Linux resources at Microsoft, including Microsoft tools that run on Linux, Linux training courses, Cloud Solution Architecture for Linux, and Microsoft + Linux news, events, and partnerships. **Microsoft Loves Linux!**